A CASE-STUDY BASED REPORT

ON

AI-BASED HELMET DETECTION FOR ROAD SAFETY USING MACHINE LEARNING

submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

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ABSTRACT

Road-safety monitoring can benefit from machine learning systems that automatically analyze images and identify visual evidence relevant to helmet compliance. This case study examines an image-based helmet dataset and documents the exploratory data analysis (EDA) carried out before model development. The analysis is intentionally image-focused because the supplied dataset is organized as image folders rather than as a conventional tabular or CSV dataset. The EDA workflow downloads the dataset, explores its folder structure, detects image-containing folders, counts images, checks every image for readability, studies image dimensions and aspect ratios, and displays representative samples. The completed analysis found 28,700 images across 287 detected folders, with 100 images in each detected folder. No corrupted or unreadable images were identified. A sample of 500 images showed a consistent resolution of 1920×1080 pixels and an aspect ratio of approximately 1.78 (16:9). These findings provide a clean and consistent starting point for subsequent machine-learning experimentation. The case study also discusses algorithm selection for a practical helmet-violation detection pipeline and the importance of responsible use when automated visual analysis is applied in road-safety settings.

1. INTRODUCTION

1.1 Background

Helmet compliance is an important visual cue in road-safety monitoring. In conventional monitoring, reviewing large volumes of traffic imagery manually can be time-consuming and difficult to scale. Machine learning and computer vision provide a way to process images systematically and support the identification of road users and helmet-related visual patterns.

This case study focuses on the data-analysis stage of an image-based helmet detection problem. The supplied EDA notebook uses the Helmet Dataset (OSF Lite) available through Kaggle and performs image-specific exploratory analysis. The notebook explicitly treats the data as an image classification dataset rather than a tabular dataset.

1.2 Motivation for the Project

The motivation is to establish whether the available image data is sufficiently organized, readable, and consistent for machine-learning experimentation. Before training a model, it is important to know how much data is available, how it is organized, whether files are corrupted, and whether image dimensions vary substantially. The EDA therefore acts as a quality-control stage before algorithm development.

1.3 Overview of the Work Carried Out

- Installed and imported the Python libraries required for image-oriented EDA.
- Authenticated with Kaggle and downloaded the dataset using kagglehub.
- Explored the directory hierarchy and identified folders containing image files.
- Counted the images associated with each detected folder.
- Created a class/folder distribution table and visualization.
- Checked all discovered images for corruption or unreadable content.
- Sampled 500 images to analyze width, height, and aspect ratio.
- Displayed representative sample images from the detected folders.
- Generated a final EDA summary containing the number of folders, images, corrupted files, and image-size statistics.

1.4 Scope and Responsible-Use Statement

The present case study is limited to dataset exploration and the machine-learning design considerations that follow from that exploration. The EDA itself does not establish a final deployment-ready enforcement system. Any real-world traffic application should use appropriate validation, human oversight, privacy protection, secure data handling, and legally authorized procedures. Model predictions should be treated as decision-support evidence rather than as an unreviewed substitute for authorized human or legal processes.

2. PROBLEM STATEMENT

2.1 Case-Study Scenario

Consider a traffic-monitoring environment in which a large number of road images must be examined for helmet-related visual information. The objective is to develop a machine-learning workflow that can learn from appropriately prepared images and eventually assist in identifying helmet compliance. The first practical

question is whether the available dataset is suitable for such experimentation and what preprocessing or modeling considerations are revealed by EDA.

2.2 Central Case-Study Question

How can exploratory analysis of the available helmet image dataset establish data quality, image consistency, and modeling requirements for a machine-learning system intended to support automated helmet-violation detection?

2.3 Formal Problem Definition

Given a collection of road-traffic images, the machine-learning task is to learn visual patterns that can distinguish helmet-related categories in images and, in a broader traffic-monitoring pipeline, support the detection of helmet violations. Formally, the system can be viewed as learning a mapping from an input image X to an output prediction Y, where Y represents the relevant helmet-related class or detection result. The exact label structure must be defined from the final training dataset used for model development; the supplied EDA notebook itself only establishes the folder-level structure of the downloaded dataset.

2.4 Objectives that Guide the Study

- Understand the structure and scale of the supplied image dataset.
- Measure the number of images available in the detected folders.
- Verify that the image files are readable and free from detectable corruption.
- Determine whether image dimensions and aspect ratios are consistent.
- Visualize the distribution of images across the detected folders.
- Inspect representative images to understand visual variability.
- Use EDA findings to identify practical preprocessing and model-design considerations.
- Establish a documented baseline for subsequent machine-learning experimentation.

2.5 Assumptions and Limitations

- The supplied notebook uses a folder-based image dataset rather than a CSV/tabular dataset.
- Folders containing image files are treated by the notebook as detected classes/folders.
- The notebook reports 287 detected folders with 100 images per folder.
- Image-size statistics are based on a random sample of 500 images rather than every image.
- The EDA does not by itself provide model accuracy, precision, recall, F1-score, or deployment performance.
- The EDA notebook does not define the final semantic helmet labels for a downstream detection model.
- Further annotation and label validation are required before supervised model training.

3. DATASET AND EXPLORATORY DATA ANALYSIS

3.1 About the Dataset

The EDA notebook identifies the source as the 'Helmet Dataset (OSF Lite)' hosted on Kaggle. The dataset is handled as an image classification-style directory structure. The notebook downloads the dataset directly in Google Colab using kagglehub after Kaggle authentication.

Dataset Attribute	Observed Value
Dataset type	Image-based dataset

Total detected folders/classes	287
Total images	28,700
Images per detected folder	100
Corrupted/unreadable images	0
Image-size sample	500 images
Mean width	1920 px
Mean height	1080 px
Aspect ratio	1.7778 (approximately 16:9)

3.2 Purpose of EDA

The purpose of EDA is to understand the dataset before model training. For an image dataset, this includes inspecting the folder organization, measuring the number of images, validating image readability, checking image dimensions, studying aspect-ratio consistency, and viewing representative samples. These checks help identify data-quality problems and inform later preprocessing and model selection.

3.3 Preparation Steps Actually Performed

- The notebook installed kagglehub and imported os, random, pathlib, NumPy, pandas, Matplotlib, PIL, and kagglehub.
- Kaggle credentials were configured through kaggle.json in the Colab environment.
- The dataset was downloaded with kagglehub.dataset_download("kronomy/helmet-dataset-by-osf-lite").
- The directory tree was traversed to understand how images were stored.
- Image extensions including JPG, JPEG, PNG, BMP, GIF, and WEBP were recognized.
- Folders containing image files were automatically detected and their image paths collected.

3.4 EDA Exploration – Implementation

3.4.1 Folder Structure and Image Counting

The notebook defines a function that recursively searches the downloaded directory. A directory is treated as a detected class folder when it directly contains supported image files. The resulting counts are stored in a pandas DataFrame and sorted in descending order.

The resulting table contains 287 rows. The displayed examples show folders such as Mandalay_1_17, Bago_highway_10, Bago_highway_17, and Bago_rural_46, each containing 100 images. This uniform count is a notable characteristic of the downloaded dataset.

3.4.2 Class/Folder Distribution Visualization

A bar chart was generated with folder name on the x-axis and number of images on the y-axis. Because the detected folders each contain 100 images, the distribution is effectively uniform at the folder level. The chart is useful as a quick verification of the counting operation.

3.4.3 Corrupted and Unreadable Image Check

Every discovered image was opened with PIL and verified. The check returned 0 corrupted or unreadable images out of 28,700 total images. This indicates that the files passed the notebook's basic image-integrity test.

3.4.4 Image Size and Aspect Ratio Analysis

A random sample of 500 images was selected for image-dimension analysis. For each readable image, the width and height were recorded and the aspect ratio was calculated as width divided by height. The descriptive statistics show no variation in the sampled dimensions: all sampled images were 1920 pixels wide and 1080 pixels high, resulting in an aspect ratio of approximately 1.7778.

3.4.5 Sample Image Inspection

The notebook also displays four randomly selected images per detected folder. This visual inspection provides a qualitative check of the image content and helps reveal differences in traffic scenes, viewpoints, and environmental conditions that may matter during model training.

3.5 Summary of EDA Findings

Finding	**Interpretation**
Dataset scale	28,700 images were discovered.
Folder structure	287 image-containing folders were detected.
Folder balance	The detected folders contain 100 images each.
Image integrity	0 corrupted/unreadable images were found.
Resolution	The 500-image sample consistently measured 1920×1080 px.
Aspect ratio	The sampled images consistently followed approximately 16:9.
Modeling implication	Consistent resolution/aspect ratio simplifies later image preprocessing, although the final training pipeline may still resize images to the model's required input dimensions.
Remaining requirement	Semantic labels/annotations for the final supervised task must be validated before training.

4. ALGORITHM SELECTION AND IMPLEMENTATION

The EDA establishes that the data is image-based and that the sampled images have a consistent 1920×1080 format. For a practical helmet-violation system, the algorithm must be capable of learning visual features from images and, where required, locating relevant objects in traffic scenes. Therefore, a computer-vision deep-learning approach is more appropriate than a purely tabular machine-learning algorithm.

4.1 Algorithm Selection Rationale

For the broader RideSafe-style helmet-violation pipeline, a YOLO-family object-detection model is an appropriate candidate because the task involves identifying visual objects and their locations in an image. The detection pipeline can be structured around motorcycle detection, occupant association, helmet-status analysis, and subsequent evidence processing. The exact final model should be selected after label compatibility, validation performance, inference speed, and hardware constraints are evaluated experimentally.

4.2 Proposed Processing Pipeline

1. Input traffic image/video frame
2. Motorcycle detection
3. Person/occupant detection and association with motorcycles
4. Occupant counting and identification of potential triple-riding cases
5. Helmet/no-helmet assessment for relevant occupants
6. Number-plate detection for confirmed violations
7. OCR of the detected number plate
8. Violation record/evidence generation

4.3 Implementation Considerations

- Resize or letterbox images according to the selected model's input requirements while preserving useful visual information.
- Use a validated train/validation/test split for model development and evaluation.
- Apply augmentation carefully so that helmets, riders, motorcycles, and plates remain visually meaningful.
- Evaluate class-wise performance instead of relying only on overall accuracy.
- For object detection, report metrics such as precision, recall, and mAP along with inference speed.
- Inspect false positives and false negatives because road scenes can contain occlusion, small objects, motion blur, and varying illumination.
- Keep the final enforcement decision subject to authorized review and applicable rules.

4.4 Expected Outcome

The expected outcome is a machine-learning workflow capable of processing traffic imagery and supporting helmet-compliance analysis. The EDA provides confidence in the basic integrity and format consistency of the examined data, while the algorithmic stage converts these prepared images into a trainable computer-vision problem. Actual model performance must be established through controlled training and evaluation on properly labeled data.

5. CONCLUSION

This case study presented an image-focused exploratory analysis of the Helmet Dataset (OSF Lite) as a preparatory stage for a machine-learning-based helmet detection problem. The analysis successfully identified

287 image-containing folders and 28,700 images. Each detected folder contained 100 images. All 28,700 images passed the notebook's readability and integrity check, with zero corrupted or unreadable files.

The image-dimension analysis of 500 randomly sampled images showed a highly consistent format of 1920×1080 pixels and an approximately 16:9 aspect ratio. This consistency is useful for designing a later preprocessing pipeline. At the same time, the EDA demonstrates an important distinction: folder structure alone does not establish the final semantic labels required for a supervised helmet-violation model. Label verification and task-specific annotation remain necessary before final training.

Overall, the EDA provides a documented, reproducible foundation for subsequent computer-vision model development. A suitable deep-learning object-detection approach can then be evaluated using properly labeled data, quantitative validation metrics, and responsible deployment practices.

APPENDIX A – EDA IMPLEMENTATION SUMMARY

The supplied Colab notebook contains the complete EDA implementation. Its main stages are dataset download, recursive folder exploration, automatic image-folder detection, image counting, class/folder distribution plotting, corruption checking, image-size/aspect-ratio analysis, sample-image visualization, and final summary printing.

- Import dependencies: os, random, pathlib, numpy, pandas, matplotlib, PIL, kagglehub.
- Download the dataset using kagglehub.
- Walk through the dataset directory and identify image-containing folders.
- Count images per folder and construct counts_df.
- Plot image count per folder.
- Open and verify every image with PIL.
- Randomly sample up to 500 images and calculate width, height, and aspect ratio.
- Display sample images and print the final EDA summary.

APPENDIX B – FINAL EDA RESULTS

Metric	Result
Total classes/folders detected	287
Total images found	28,700
Corrupted images	0
Sampled images for dimension statistics	500
Minimum image width in sample	1920 px
Maximum image width in sample	1920 px
Minimum image height in sample	1080 px
Maximum image height in sample	1080 px
Aspect ratio	1.777777...

Signature of the Team members